

The Potential of Google NotebookLM for Teaching and Learning

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Abstract: Google NotebookLM is an AI-powered tool that allows users to upload documents and interact with them through grounded summarisation, contextual question answering, and synthesis. This paper examines its applications in higher education across research, study, teaching, and courseware development. As a research tool, it can streamline literature reviews, highlight key themes, and ensure citation accuracy. For students, it functions as a personalised tutor by creating flashcards, summarising notes, and clarifying complex concepts from course materials. Educators can use it to draft lesson plans, design quizzes, and efficiently structure online content. In teaching delivery, NotebookLM could support adaptive learning by addressing individual student queries and identifying misconceptions. Its grounded approach can reduce hallucination risks common to generative AI, thereby promoting academic integrity and confidence in AI-assisted learning. By combining document-based responses with conversational interaction, NotebookLM can enhance cognitive engagement, accommodate diverse learning preferences, and facilitate more flexible, self-directed study. This paper offers a critical overview of NotebookLM's educational and research potential, highlighting important considerations for its responsible integration into higher education.

Keywords: NotebookLM, generative AI, personalised learning, study tools, educational technology, higher education, grounded AI, teaching innovation.

Introduction

The integration of generative artificial intelligence (GenAI) into higher education is transforming traditional teaching methods, encouraging educators to explore new approaches to content delivery, assessment, and student support. While tools like ChatGPT have garnered attention for their capabilities, concerns persist about their potential to generate inaccurate or misleading information. In response, platforms such as Google NotebookLM have appeared, providing document-based AI interactions that focus on accuracy and context (Google, 2024).

NotebookLM enables users to upload documents, such as lecture notes, research articles, and transcripts, and interact with them through AI-driven summarisation, question answering, and synthesis. This personalised method encourages deeper engagement with content, helping learners create study aids such as mind maps and timelines, and assisting educators in streamlining lesson planning and assessment development (Hu, 2024). By grounding responses in user-provided materials, NotebookLM reduces the risk of AI-generated misinformation, thereby enhancing academic integrity and fostering trust in AI-supported learning environments.

The adoption of GenAI tools, such as NotebookLM, aligns with the drive towards innovative, student-centred learning environments. Embedding GenAI into curricula is crucial for fostering critical thinking and adaptability among students (Reyna, 2025). Additionally, integrating such tools requires a reevaluation of assessment strategies to ensure meaningful learning outcomes and uphold academic standards (Reyna, 2023).

This paper examines the diverse applications of Google NotebookLM in higher education, highlighting its potential as a research assistant, study aid, and tool for developing course content. As NotebookLM continues to develop, this paper provides a snapshot in time, analysing its current features and effects to offer insights into how document-based GenAI can improve personalised learning and foster pedagogical innovation.

Advantages of Google NotebookLM for Teaching and Learning

Google NotebookLM offers several pedagogical and practical advantages in higher education contexts, especially when compared to traditional large language models (LLMs) such as ChatGPT or Copilot. A core advantage lies in its retrieval-augmented generation (RAG) architecture, which grounds outputs strictly in user-uploaded source documents. This design significantly reduces hallucination, which is the generation of inaccurate or fabricated information and enhances the reliability and transparency of AI responses (Tozuka et al., 2024; Johnno et al., 2024).

Unlike general-purpose LLMs, NotebookLM enables learners and educators to interact directly with personalised content, such as lecture slides, readings, or transcripts, promoting more relevant and contextualised learning experiences. In a study exploring its use in physics education, Tufiño (2024) found that NotebookLM effectively supported Socratic-style dialogue and active learning by referencing teacher-curated materials. This personalised interaction model fosters deeper engagement and critical thinking, aligning with constructivist pedagogical principles.

Furthermore, NotebookLM supports multi-modal outputs, including mind maps, timelines, FAQs, and podcast-style summaries, allowing learners to review information in formats that match their cognitive and sensory preferences (Hu, 2024). Such features may benefit neurodiverse learners or those with varied learning preferences by enhancing accessibility and reducing cognitive load.

The tool also has substantial implications for academic integrity and trust, as all outputs include citations linked back to the user-provided sources. This transparency addresses ongoing concerns about the ethical use of GenAI in education and supports the responsible adoption of AI (Reuter et al., 2024). Moreover, educators can use NotebookLM to streamline online content development, such as generating quiz questions, summarising key readings, and developing formative assessments, thereby saving time while ensuring content alignment with intended learning outcomes (ED Technology Specialists, 2024). These features position it as a valuable tool not only for learners seeking a more profound understanding but also for educators aiming to innovate their teaching with AI responsibly and effectively.

Drawbacks and Considerations of Using Google NotebookLM in Higher Education

While Google NotebookLM presents several pedagogical advantages, its adoption in educational contexts also brings important challenges and risks that must be carefully considered. One big concern is copyright and intellectual property. NotebookLM now lets users publicly share notebooks (via links) with others, much like Google Docs (The Verge, 2025). However, this raises a significant copyright issue: if a notebook includes journal articles, book chapters, or other copyrighted content, sharing it publicly likely infringes copyright law. It may also violate Google's terms of service, unless one of the following conditions is met:

- The user has full rights to distribute the uploaded materials,
- The notebook is shared in 'chat only' mode without embedding the original content, or
- A licensing agreement explicitly permits redistribution.

Another key drawback is cost and scalability. While NotebookLM can be used for free with limitations, Google has introduced NotebookLM Pro, a premium tier with expanded features (Steele.com, 2025). In Australia, the monthly cost is approximately US\$20/month. For students, the cost is US\$10/month. This raises concerns about equity and access, particularly for students or institutions that lack the financial resources to adopt paid tools. The emergence of a tiered model could further widen the digital divide in already resource-constrained educational environments (Selwyn, 2023).

There are also technical limitations and risks of overreliance. Although NotebookLM minimises hallucinations by grounding answers in user-provided content, users may still misinterpret the scope or authority of AI-generated responses, particularly if documents themselves contain errors or outdated information. Additionally, users may upload documents that contain inaccuracies or biases, which can be problematic. Tufiño (2024) noted that

while the tool performed well as a physics tutor, it still required educator oversight to ensure pedagogical accuracy. Moreover, since the tool only generates responses based on uploaded content, learners could develop a false sense of comprehensiveness, unaware of important knowledge gaps beyond the curated sources.

Concerns exist regarding data privacy and institutional governance. Although Google states that uploaded content remains private and secure, institutions may need to assess compliance with local data protection laws, particularly when using the platform in regulated environments or involving sensitive student data.

Lastly, there is the risk of platform discontinuation, a pattern evident in Google's history of retired services, such as Google Jamboard (discontinued in January 2025), Google Wave (2010–2012), and Google Buzz (2010–2011), despite their widespread adoption (Guinness, 2023). These precedents create uncertainty about the long-term availability of NotebookLM, which could significantly disrupt teaching and research workflows if service support is withdrawn.

While NotebookLM holds promise as a contextualised, AI-powered educational tool, its implementation must be guided by robust policy, critical digital literacy, and clear ethical frameworks to avoid legal, pedagogical, and accessibility pitfalls.

Google NotebookLM Overview

NotebookLM is a context-aware generative AI application that enables users to create a 'notebook' by uploading documents, including PDFs, lecture notes, or research articles. The AI then enables interaction with this content through summarisation, question answering, and synthesis, making it a valuable educational assistant that adapts to user-provided information. The following section discusses its application in higher educational settings as a research companion, study tool, courseware development, curriculum review and renewal, and teaching delivery.

As a Research Companion

Google NotebookLM offers significant potential as a research companion in higher education by enhancing how students and academics interact with source material. Through its retrieval-augmented generation (RAG) architecture, NotebookLM bases its responses exclusively on user-uploaded documents, such as journal articles, reports, lecture notes, and datasets, enabling precise and contextually relevant interactions (Tozuka et al., 2024).

One of its most impactful features for academic research is the automation of literature reviews. By summarising uploaded texts, clustering similar concepts, and identifying overarching themes, NotebookLM could help researchers quickly gain insight into large volumes of information. This accelerates the early stages of inquiry while reducing the cognitive load often associated with synthesising fragmented sources (Tufiño, 2024).

NotebookLM also supports citation-aware question answering, meaning users can query across multiple documents and receive direct answers with inline references pointing back to the source material. This reinforces academic integrity by making it easier to trace ideas, support claims, and avoid accidental plagiarism, common concerns in GenAI-supported writing tasks (Reyna, 2023).

Additionally, by enabling cross-document synthesis and insight generation, the tool facilitates interdisciplinary research and could help novice researchers develop stronger information literacy skills. Its ability to deliver answers in different formats, such as timelines, FAQs or podcasts, makes it a versatile tool not only for individual scholarship but also for collaborative academic environments and supervision settings.

As research becomes increasingly data-intensive and time-constrained, tools like NotebookLM offer a scalable way to augment human inquiry without replacing the critical thinking and methodological rigour central to academic work.

As a Study Tool

Google NotebookLM could have strong potential as a personalised study companion, supporting diverse learning needs and promoting self-regulated learning practices. By leveraging uploaded course materials (e.g., lecture notes, textbooks, readings, and transcripts), NotebookLM could enable students to interact with content in meaningful ways tailored to their individual study preferences.

One key feature is its ability to generate personalised study aids, including summaries, FAQs, timelines, podcasts, and flashcards. These AI-generated resources could help learners organise, review, and consolidate knowledge without relying on generic or pre-scripted explanations. This capability is particularly beneficial for exam preparation and topic revision, where reducing cognitive load and efficiently encoding information are crucial (Hu, 2024).

The podcast feature could be a transformative addition to AI-assisted learning. NotebookLM can convert complex scientific papers into engaging audio dialogues between two virtual speakers who explain and unpack the content conversationally. Moreover, users can join the podcast in real time, ask questions, and extend the discussion dynamically. This interactive, multimodal approach has strong pedagogical potential, especially for auditory learners or those seeking deeper conceptual understanding through dialogue and repetition.

NotebookLM could also function as a real-time, context-aware tutor by answering questions grounded in the materials the student has uploaded. Unlike open-ended LLMs that draw from broad internet sources, NotebookLM tailors its responses exclusively to the learner's documents, ensuring content relevance, terminology alignment, and instructional consistency. This could support a more profound understanding and reduce the risk of confusion from off-topic or inaccurate explanations (Tufiño, 2024).

Importantly, this interaction could encourage self-regulated learning, where students take an active role in planning, monitoring, and evaluating their learning. GenAI tools that promote learner agency and reflection, rather than passive content consumption, align with future-oriented educational goals (Reyna, 2025). NotebookLM enables students to ask their questions, review AI responses, and assess their understanding independently, creating opportunities for metacognitive engagement and iterative learning.

By integrating pedagogical soundness with AI-enhanced flexibility, NotebookLM could offer a scalable and inclusive tool to meet the evolving demands of higher education learners.

For Courseware Development

Google NotebookLM offers promising applications for educators involved in the design and delivery of courseware content. As demands for scalable, high-quality digital resources continue to grow, NotebookLM could provide an AI-assisted workflow that enables educators to scaffold content, draft assessments, and rapidly develop instructional materials, all while ensuring alignment with course objectives and curriculum standards.

By uploading unit outlines, lesson plans, and reference texts, educators can use NotebookLM to generate instructional scripts, learning guides, and even formative assessments such as quizzes and discussion prompts. These outputs are grounded in source materials, which enhances curricular coherence and reduces the time typically spent writing and revising manually. Tufiño (2024) demonstrates that NotebookLM can generate Socratic-style explanations and questions using pre-curated content, making it suitable for constructing interactive learning experiences.

The tool also assists with scenario development for problem-based or case-based learning by synthesising multiple documents to create contextually rich narratives. This is particularly valuable in clinical, business, or law education, where scenarios must accurately reflect domain-specific details. As the author argues in a previous paper (Reyna, 2025), AI tools that can generate draft content aligned with learning outcomes have the potential to accelerate learning design while preserving pedagogical integrity.

Moreover, NotebookLM supports content modularity and reusability, enabling educators to quickly adapt material for different delivery modes (e.g., online, hybrid, self-paced). Its ability to summarise and restructure existing resources into new formats allows for greater agility in content production, which is crucial in rapidly evolving subject areas or during times of curriculum reform.

In essence, NotebookLM acts as a pedagogically sensitive co-designer, allowing educators to retain creative and academic control while benefiting from the efficiency and customisation offered by GenAI.

Curriculum review and renewal

Curriculum review and renewal are essential processes in higher education, ensuring that program design remains relevant, coherent, and aligned with institutional goals, disciplinary standards, and graduate attributes. Traditionally, these processes are time-consuming and labour-intensive, requiring careful analysis of course materials, mapping of learning outcomes, and verification of alignment with program-level objectives. Google NotebookLM could support and streamline these efforts by offering AI-assisted content analysis and synthesis grounded in institutionally curated documents.

By uploading unit outlines, course descriptors, assessment tasks, and program learning outcomes, academic teams could use NotebookLM to perform rapid content audits. The tool's summarisation and question-answering features may assist in identifying redundancies, inconsistencies, or gaps across units. For example, NotebookLM could be prompted to trace where specific graduate attributes, such as critical thinking, communication, or ethical reasoning, are addressed within a program. Similarly, it could help verify whether assessment tasks meaningfully align with stated learning outcomes or where additional scaffolding may be required.

Moreover, NotebookLM's retrieval-augmented generation capabilities could facilitate constructive alignment checks by analysing whether the intended learning outcomes (ILOs), teaching activities, and assessment tasks across different courses are pedagogically coherent (Biggs & Tang, 2011). It could also help surface thematic trends or misalignments in large, multi-unit programs, supporting strategic curriculum planning.

During curriculum renewal phases, NotebookLM could serve as a collaborative tool for course design teams by generating draft curriculum maps, summarising external benchmarking documents (e.g., accreditation standards or professional frameworks), and synthesising stakeholder feedback documents. This would enhance transparency and traceability of curriculum decisions, helping teams make evidence-informed updates that reflect evolving educational and industry expectations.

Importantly, NotebookLM's ability to work with institution-specific documentation could ensure that curriculum review remains contextually grounded, avoiding the pitfalls of generic AI-generated outputs. As curriculum alignment with graduate attributes becomes an increasing focus for quality assurance and accreditation, tools like NotebookLM could provide a scalable, structured, and efficient method for academic teams to engage in meaningful curriculum development processes.

In Teaching Delivery

Beyond content development and study support, Google NotebookLM also shows considerable promise in delivering teaching, particularly in enabling more responsive, efficient, and personalised classroom experiences. By interacting with uploaded course content, NotebookLM could support educators during preparation, instruction, and formative assessment.

One of its most immediate applications is in lecture preparation. Educators can upload lecture notes, reading materials, or prior presentation slides, and use NotebookLM to generate summaries, FAQs, illustrative examples, or even opening prompts that align with learning objectives. This not only saves time but also helps ensure instructional consistency and clarity, especially in courses taught by multiple instructors or across different campuses.

NotebookLM could also facilitate real-time clarification of student questions by providing AI-supported answers based on course-specific documents. During office hours, discussion forums, or online sessions, students can ask clarifying questions and receive relevant responses grounded in their course materials, minimising the risk of misinformation and easing the educator's workload. As demonstrated by Tufiño (2024), NotebookLM is particularly useful in managing high-volume Q&A settings, functioning as a contextual tutor that echoes the course's terminology and pedagogy.

Additionally, the tool's dialogue-based interaction can help identify student misconceptions and gaps in understanding. When students consistently ask questions about the same topics or misunderstand specific concepts, educators can use NotebookLM transcripts to inform targeted reteaching strategies or develop supplementary content.

Perhaps most significantly, NotebookLM could enable a more personalised approach to instruction. By allowing students to engage with content at their own pace and in formats that match their cognitive preferences, such as audio explanations, timelines, or scaffolded answers, educators can more effectively support diverse learners, including those in asynchronous or hybrid learning environments.

When integrated thoughtfully, NotebookLM can enhance not only the efficiency of teaching but also the responsiveness and inclusiveness of instructional delivery.

Reimagining Library Services with GenAI

The emergence of document-grounded AI tools like Google NotebookLM signals a transformative opportunity for university libraries to reimagine their digital services. While current library systems offer powerful search and citation management tools, they often fall short of supporting deeper, AI-assisted engagement with scholarly literature. Developing an institution-specific platform with capabilities like NotebookLM, integrated within library databases, could mark a new era in academic research and information literacy.

Such a platform could enable students and academics to search institutional and open-access databases, add selected articles directly to personalised notebooks, and interact with this content through AI-powered summarisation, question answering, and synthesis. This functionality would not only streamline literature reviews and support the development of systematic or scoping reviews but also help users identify conceptual gaps, emerging themes, or conflicting findings within a body of literature. By grounding all AI-generated outputs in institutionally licensed or open-access content, the platform would address major concerns around hallucination and misinformation, which are prevalent in general-purpose large language models.

Moreover, embedding this functionality within the academic library ecosystem could foster critical thinking, problem-solving, and GenAI literacy. Students would engage more actively with research material, developing the ability to interrogate sources, evaluate claims, and construct knowledge iteratively. Academics, in turn, could benefit from AI-supported synthesis that reduces time spent navigating fragmented databases and enhances interdisciplinary exploration.

From a pedagogical perspective, such a tool would extend the role of the academic library from a repository of resources to an active, intelligent partner in the learning and research process. It would also align with the broader institutional mission of preparing students for a data-rich, AI-augmented future, where the ability to engage with information critically is as important as access to it.

Developing AI-enhanced research environments within university libraries presents both a challenge and an opportunity. It requires thoughtful collaboration among librarians, academic staff, IT services, educational technologists, and learning designers. However, the potential to enhance scholarly efficiency, transparency, and innovation makes it a compelling direction for future-ready institutions.

Fostering GenAI Literacy Through NotebookLM

Integrating Google NotebookLM into higher education does more than enhance academic outcomes; it can cultivate GenAI literacy, a critical skill set for the future workforce. GenAI literacy involves understanding, evaluating, and using AI tools ethically, critically, and creatively (Holmes et al., 2022). By engaging with NotebookLM, students can develop habits of inquiry grounded in source-based reasoning, contextual awareness, and reflective questioning, skills essential not only in academic contexts but also across professional and personal domains.

NotebookLM's grounding in user-provided materials trains students to critically evaluate the reliability of outputs, reinforcing evidence-based thinking and responsible tool use. These competencies are transferable to everyday challenges. For example, in health sciences, learners can synthesise dietary guidelines and well-being frameworks to create personal health plans. In cross-cultural engagement, uploading cultural safety resources and local protocols can enhance responsiveness and communication. Even outside formal education, users can curate documents for various purposes, such as travel, budgeting, buying a property, creative writing, or civic decision-making, with each interaction reinforcing AI literacy through contextually rich content.

Rather than treating AI as a passive assistant, NotebookLM could encourage active learning and informed judgment. Preparing students for the future requires more than tool proficiency; it demands the ability to critically evaluate, personalise, and ethically apply AI (Reyna, 2025). NotebookLM offers a practical and accessible approach to integrating these capabilities into everyday learning, equipping students for lifelong digital adaptability.

Conclusion

Google NotebookLM could offer a flexible, grounded, and user-driven approach to generative AI in education. By enabling interactions with user-provided documents, it has the potential to overcome key limitations of general-purpose AI tools, particularly hallucination and lack of contextual relevance. Its ability to support research synthesis, generate personalised study aids, scaffold courseware materials, and enhance real-time teaching delivery makes it a valuable tool across the academic lifecycle. Furthermore, its integration into higher education could serve as an authentic platform for fostering GenAI literacy, equipping students for critical, ethical, and lifelong engagement with AI technologies.

However, these developments remain in their early stages, and the body of evidence (including empirical and pilot studies) surrounding NotebookLM's educational impact is still emerging. Further research is needed to explore how tools like NotebookLM can be effectively integrated within higher education ecosystems, not only from a technological perspective, but also through the lenses of institutional systems, academic practices, and student learning experiences. Considerations around data governance, accessibility, and intellectual property must be addressed proactively, alongside professional development for educators and digital literacy initiatives for learners. As institutions navigate this evolving landscape, the integration of AI must be grounded in pedagogical principles and driven by empirical evidence to ensure its value is both sustainable and transformative.

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